

hcv
Home control Modul for Viessmann Vitocal

Please also read the HCV Description in the docu folder.

Local Viessmann E3 CAN-to-MQTT Bridge

An isolated, passive CAN-bus adapter to fetch local data from Viessmann E3-generation heat pumps (e.g., Vitocal 250-A, 252-A) and stream it via Wi-Fi to MQTT with native Home Assistant Auto-Discovery.

🚀 Key Features

- **100% Local Control:** No internet dependency, no official ViCare Cloud API rate limits.
- **Hardware Protection:** Features full galvanic isolation to protect your expensive heat pump from electrical damage.
- **Passive Sniffing:** Read-only operation ensures zero interference with the heating system's warranty.
- **Plug & Play for Home Assistant:** Built-in MQTT Discovery automatically creates all entities (temperatures, flow rates in L/h, energy in kWh, and states) without writing a single line of YAML.
- **Universal Compatibility:** Works seamlessly with Home Assistant, ioBroker, Node-RED, openHAB, and Symcon.

🛠️ Project Structure & License Model

This project follows a hybrid **Open Hardware / Commercial Software** model to support the community while funding the extensive reverse-engineering work behind the Viessmann CAN protocols.



1. Open Hardware (Free)

All hardware files are open-source. You can find the schematics, PCB layout files (KiCAD/EasyEDA), Bill of Materials (BOM), and a 3D-printable enclosure STL file in the ``/hardware`` folder. Feel free to DIY and order your own boards.

2. Pre-Assembled Hardware (Optional)

If you do not want to solder SMD components or lack the equipment, you can buy a fully assembled and tested hardware unit inside its enclosure directly from me for **approx. €35** (plus shipping from Germany).

3. Firmware & Software License

- **Demo Version (Free):** Includes the local Wi-Fi/MQTT captive portal captive setup, hardware connection test, and displays your unique ESP32 Chip ID.
- **Full Version (€50):** Unlocks all heating data points, daily energy counters, and native Home Assistant Auto-Discovery. It is a one-time payment per device and works **100% offline** (via a license file uploaded to the file system).

📁 Getting Started

Hardware Connection

Connect the adapter to the external CAN-bus of your Viessmann heat pump (typically labeled as "Plug 91" / Stecker 91 on the main terminal block).

Firmware Setup

1. Flash the free firmware build to your ESP32.
2. Power on the device and connect to the Wi-Fi Access Point generated by the adapter.
3. Open `http://192.168.4.1` in your browser, enter your local Wi-Fi and MQTT broker credentials.
4. Copy the unique **Chip ID** shown on the web interface.

Activating the Full Version

To unlock all sensors and Home Assistant Auto-Discovery:

1. Reach out to me via **PM** in the **Forums** or via Email at `info@embdes.com` with your **Chip ID**.
2. After payment (€50), you will receive a `license.txt` file.
3. Upload this file through the adapter's web interface and reboot the device.
4. Open Home Assistant → **Settings** → **Devices & Services** → **MQTT**—your heat pump will show up automatically!

📊 Supported Data Points (Extract)

- **Analogue Sensors:** Flow/Return temperatures, Outdoor temperature, Domestic Hot Water (DHW) temp, Compressor frequency.
- **Flowrates:** Allengra Sensor / Volumetric flow rate (properly formatted in `L/h`).
- **Energy Dashboard:** Daily energy consumption (properly formatted in `kWh` with `total\_increasing` state class).
- **Binary States:** Compressor status, Mixer pump status, Error flags (formatted with native `running` and `problem` device classes).

Sample of ioBroker Integration

mqtt		ioBroker	
0		1054	
1054		EVC	mqtt client variable
EVC		HCV	mqtt client variable
HCV			
vcal		vcal	mqtt client variable
AllengraSensor		HCV/vcal/AllengraSensor	mqtt client variable
BackendConnectionStatus		HCV/vcal/BackendConnect	mqtt client variable
CentralHeatingPump		HCV/vcal/CentralHeatingP	mqtt client variable
CompressorInletPressureSensor		HCV/vcal/CompressorInlet	mqtt client variable
CompressorInletTemperatureSensor		HCV/vcal/CompressorInlet	mqtt client variable
CompressorOilTemperatureSensor		HCV/vcal/CompressorOilTe	mqtt client variable
CompressorOutletPressureSensor		HCV/vcal/CompressorOutl	mqtt client variable
CompressorOutletTemperatureSensor		HCV/vcal/CompressorOutl	mqtt client variable
CompressorSpeedPercent		HCV/vcal/CompressorSpee	mqtt client variable
CurrentThermalCapacitySystem		HCV/vcal/CurrentThermalC	mqtt client variable
DomesticHotWaterPump		HCV/vcal/DomesticHotWat	mqtt client variable
DomesticHotWaterSensor		HCV/vcal/DomesticHotWat	mqtt client variable
EconomizerLiquidTemperatureSensor		HCV/vcal/EconomizerLiqui	mqtt client variable

Contact & Support

If you want to buy a pre-assembled board or a software license, please contact me:

- ****Home Assistant Forum:**** [[@chrkir](https://home-assistant.io)](https://home-assistant.io)
- ****ioBroker Forum:**** [[@chrkir](https://iobroker.net)](https://iobroker.net)
- ****Email:**** ``info@embdes.com``

***Disclaimer:** This is a third-party community project. This project is not affiliated with, authorized, or endorsed by Viessmann.*